

NORSE

MERCURY AND CYANIDE IN SMALL-
SCALE GOLD MINING

What studies from Ghana, Uganda and Tanzania show about recovery and process selection

Basis for the NORSE process philosophy: gravity first, concentration before chemistry, and mineralogy-led process selection.

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1. Executive Summary

Mercury and cyanide have become standard tools across much of artisanal and small-scale gold mining (ASGM), but they are used for different reasons and solve different process problems. Mercury provides a simple and rapid route from a gravity concentrate to a visible gold product. Simple VAT cyanidation has, at the same time, become a common method for recovering gold from tailings left after earlier gravity and amalgamation stages.

The reviewed studies show that these processes are not necessarily metallurgically optimal. In Ghana, direct smelting of a well-upgraded gravity concentrate achieved 98.3% recovery in field trials, compared with 88% for amalgamation. In Uganda, the mercury-free direct-smelting method produced 1.4 g of gold compared with 1.0 g using amalgamation in the comparative test. [1][2]

Tanzanian data also show that small-scale VAT cyanidation often operates with significant residual gold losses. A study from Nyarugusu, Geita reported existing recovery of 60-70%, which could be improved to approximately 83% after optimisation of particle size, pH and cyanide concentration. A 2023 national review describes VAT leaching as established within ASGM while also pointing to inefficient processing practices. [3][4]

At Mpanda, mineralogical analyses showed why adding more cyanide did not automatically improve results: reactive copper minerals and carbonaceous material interfered with the process. Under controlled laboratory conditions, 89.4% recovery was achieved with ammoniacal thiosulphate, underlining that mineralogy should guide process selection. [5]

Key Findings

Location / study	Process comparison	Result	Main conclusion
Ghana	Direct smelting vs mercury	98.3% vs 88% field	A high-grade gravity concentrate can be smelted directly.
Uganda	Direct smelting vs mercury	1.4 g vs 1.0 g	The mercury-free method produced more gold in the test.
Nyarugusu, Geita	VAT cyanidation	60-70% → 83% after optimisation	Process control has a major impact on recovery.
Mpanda	Cyanide vs thiosulphate testwork	89.4% with thiosulphate	Mineralogy can make standard cyanidation unsuitable.

NORSE process principle: Recover physically first. Upgrade the concentrate. Smelt directly where appropriate. Analyse the remainder, then let mineralogy determine whether any leaching step is justified.

2. Background and Context

Mercury has been used in gold extraction for centuries. Within ASGM, the method has persisted because it is simple, fast, requires limited equipment and provides an immediate, visible result. When a method works well enough and infrastructure for alternatives is lacking, it is passed from miner to miner without necessarily being reassessed from a metallurgical perspective.

Cyanide serves a different function. It dissolves gold chemically and can therefore recover gold that was not captured in earlier gravity or amalgamation stages. In north-western Tanzania, cyanide plants have created a market for tailings that were previously regarded as waste. IPIS estimated around 200 cyanide plants in the four mapped regions, while emphasising that exact figures are unavailable because many facilities operate informally. [6]

However, the reviewed sources do not provide sufficient evidence to claim that exactly 95% of all leaching plants in Tanzania are VAT operations. This report therefore uses the more defensible formulation that VAT leaching is common and well established in small-scale gold processing in Tanzania.

3. Scope and Method

This report is a targeted literature review focusing on four questions:

- Whether a high-grade gravity concentrate can be smelted directly without mercury while maintaining or improving recovery.
- What actual recovery has been reported from small-scale VAT cyanidation in Tanzania.
- Which mineralogical factors can explain low recovery in standardised cyanide treatment.
- What process implications the findings have for NORSE and NMC.

The source base consists of peer-reviewed articles, university- or institution-published versions of articles, and an extensive field mapping of ASM in north-western Tanzania. The results should not be interpreted as universal recovery figures for all ores; they show what was achieved under the specific mineralogical and operating conditions investigated in each study.

4. Findings

4.1 Why mercury persists

Mercury's main strength in ASGM is practical. It binds free gold into an amalgam that can be readily separated from heavy mineral sand. The miner can then heat the amalgam and obtain a visible gold product without advanced laboratory or process infrastructure.

This does not mean, however, that amalgamation is technically the best recovery method. On the contrary, the Ghana and Uganda studies show that when a sufficiently clean and high-grade gravity concentrate is first produced, direct smelting can achieve at least comparable and, in the cases studied, better results.

4.2 Ghana: direct smelting versus amalgamation

Amankwah et al. tested direct smelting as an alternative to amalgamation in small-scale gold mining in Ghana. Gravity concentrates were upgraded before smelting. In the laboratory, direct smelting achieved 99.8% recovery compared with 97% for amalgamation. In the field, a locally manufactured furnace, the sika bukyia, was used, and average recovery was 98.3% compared with 88% for amalgamation. [1]

The important point is the process sequence. Direct smelting does not replace gravity separation. The method requires a small and sufficiently high-grade concentrate to be produced first. The study therefore supports a flowsheet in which gravity is followed by cleaning/upgrading and then direct smelting, rather than automatically inserting a mercury step.

4.3 Uganda: direct smelting versus amalgamation

In Tiira, Uganda, a practical comparison was conducted between the local amalgamation method and a mercury-free method in which the concentrate was smelted with borax using a Filipino method. Standardised quantities of ore of comparable quality were used. [2]

Method	Mercury	Gold product	Time
Amalgamation	4 g	1.0 g	53 min
Mercury-free direct smelting	0 g	1.4 g	62 min

The mercury-free process produced 40% more gold in this particular test but took nine minutes longer. The study was limited, and the 40% improvement should not be used as a general recovery factor. The robust conclusion is instead that mercury is not necessary to produce a saleable gold product when the gravity concentrate can be properly upgraded and smelted.

4.4 Tanzania: VAT cyanidation in small-scale processing

VAT leaching is attractive for small-scale operations because the plant can be built relatively simply and without the full mechanical agitation used in, for example, CIL/CIP systems. One important application in Tanzania is the treatment of gold-bearing tailings from earlier gravity processes.

Maganga et al. (2023) describe processing practices in the Tanzanian ASGM sector and conclude that inefficient methods remain in use. The study also presents a mercury-free flowsheet as a basis for further development. [4]

The IPIS field mapping shows the economic logic behind the cyanide industry: when the first process leaves substantial gold in the tailings, a commercial value arises in purchasing and leaching that material. This also means that the tailings market itself is indirect evidence of inefficiency in the first recovery chain. [6]

4.5 Nyarugusu, Geita

Hiji and Maganga studied small-scale VAT leaching at Nyarugusu in Geita. The existing processes were reported to achieve approximately 60-70% recovery, despite cyanide concentrations of up to about 1,200 ppm, pH of around 10.5-12, particle sizes of around 300-360 μm , and 6-7 days of leaching without agitation. [3]

Using Response Surface Methodology, three key parameters were optimised. The highest reported recovery, approximately 83%, was achieved at about 950 ppm cyanide, pH 10.5 and a particle size of 240 μm . The result shows that more reagent does not automatically produce better recovery; particle size, contact, pH and process control are critical.

4.6 Mpanda: when mineralogy defeats a standard flowsheet

At the Mpanda Mineral Field, the cause of persistently low gold recovery was investigated. Mineralogical and chemical characterisation identified reactive copper minerals - including covellite, cuprite and chalcocite - as well as metallic copper and carbonaceous material. [5]

These minerals can interfere with cyanidation through reagent consumption and preg-robbing. The study therefore tested ammoniacal thiosulphate and achieved a maximum recovery of 89.4% under optimised laboratory conditions. The result should not be generalised to all Tanzanian ores, but it demonstrates why process selection must be based on mineralogy and metallurgical testwork rather than tradition.

5. Analysis and Implications

The reviewed studies point to a common structural problem: process chains within ASGM have often evolved historically, step by step, to solve immediate practical problems. They are therefore not necessarily designed as coherent metallurgical flowsheets.

Traditional Process Chain Compared with the NORSE Principle

Traditional process logic	NORSE process logic
Crushing → Grinding → Gravity → Mercury → Tailings → VAT cyanide	Crushing → Grinding → Gravity → Cleaning → Upgrading → Direct Smelting
Chemicals are used because earlier stages have left gold behind or failed to separate it efficiently.	Recover physically first. Then analyse the residual fraction and select any chemical treatment according to mineralogy.

For NORSE, the strategically most important conclusion is therefore not that cyanide should always be prohibited or that one particular alternative chemical is always better. Such a position would be technically weak. The stronger position is that chemical treatment should not be the default before mineralogical understanding and testwork.

This leads to a clear NMC logic: a site-based gravity stage maximises physical recovery and produces an upgraded concentrate. The concentrate can proceed to the Gold Room and direct smelting where metallurgically appropriate. Residual material that still contains economically relevant gold is analysed separately. Only then is it determined whether further treatment - for example thiosulphate or another process - is justified.

6. Recommendations

Priority	Recommendation
A - Implement now	Make Gravity → Cleaning → Upgrading → Direct Smelting the standard baseline for free-milling material when testwork shows that the concentrate is suitable.
A - Implement now	Introduce basic mineralogical screening and bottle-roll/leach testwork before sizing any leaching process for a new PML.
A - Document	Use the Ghana and Uganda studies as evidence that mercury-free direct smelting can be both technically and economically relevant when the concentrate is sufficiently upgraded.
B - Verify Tanzania	Map 5-10 representative VAT plants in Kahama/Geita and measure head grade, tail grade, cyanide consumption, retention time and actual recovery.
B - Verify claim	Do not use the claim that 95% of Tanzania's leaching plants are VAT operations until a verifiable national source is identified.
B - Pilot	Conduct a NORSE side-by-side test: the existing local process versus NORSE gravity/cleaning/direct smelting, with mass balance and assays on all streams.
C - Selective chemistry	Retain thiosulphate as a selective option for suitable mineralogy, not as a universal replacement for cyanide.

7. Risks and Limitations

- Recovery figures are ore-specific and must not be transferred directly between deposits.
- The Uganda comparison was a limited practical test, and the 40% improvement should not be used as a general design factor.
- The Ghana results require a sufficiently high-grade and well-cleaned gravity concentrate.
- VAT performance in Tanzania varies with mineralogy, particle size, oxygen transfer, pH, cyanide concentration, retention time and operator control.
- The Mpanda result demonstrates the potential of thiosulphate for the specific mineralogy studied; it is not a general argument for thiosulphate for all ores.
- National statistics on the number and type of small-scale leaching plants are incomplete, particularly for informal operations.

8. Conclusion

The central conclusion is that framing the choice as mercury versus cyanide is too narrow. Both have become standard tools because they solve practical problems in process chains where concentration, mineralogical understanding and process control are often limited.

The Ghana and Uganda studies show that a well-upgraded gravity concentrate can proceed directly to smelting without mercury and with very high recovery. Tanzanian VAT studies also show that cyanide in itself does not guarantee high recovery. Nyarugusu demonstrates the importance of process control; Mpanda demonstrates the importance of mineralogy.

The correct process question is therefore not "mercury or cyanide?" but: What does this ore actually require?

For NORSE, this means: physical separation first, direct smelting where appropriate, and chemistry only after mineralogy and metallurgical testwork justify it.

9. References

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Appendix A - Evidence summary

Source	Material / setting	Process	Reported outcome	Interpretation	NORSE relevance
Ghana 2010	Gravity concentrate	Direct smelting vs amalgamation	98.3% vs 88% field	Direct smelting can outperform mercury after adequate upgrading.	Supports gravity-cleaning-smelting baseline.
Uganda 2019	Tiira ASGM ore	Borax direct smelting vs mercury	1.4 g vs 1.0 g	Mercury-free route produced more gold in the test.	Strong demonstration / training case.
Nyarugusu 2015	Small-scale VAT	Cyanide VAT	60-70% existing; 83% optimised	Operating parameters matter substantially.	Do not equate cyanide use with high recovery.
Tanzania 2023	ASGM national review	Multiple practices	Inefficient methods remain common	System-level process improvement opportunity.	Supports NMC demonstration/testing role.
Mpanda 2023/24	Cu-bearing, carbonaceous ore	Cyanide problem; thiosulphate test	89.4% optimised thiosulphate	Mineralogy controls process response.	Supports mineralogy-led selection.
IPIS 2019	447 mapped ASM sites	Tailings + cyanide economy	~200 cyanide plants estimated in 4 regions	Tailings have become a secondary ore stream.	Shows economic cost of poor first-pass recovery.